Algorithmic Collusion via Meta-Learning in Multi-Agent Environments

Dissertation Introduction & Context

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"Can AI agents learn to fix prices without being told to?"

1. What is Machine Learning (ML)?

The Formal Definition: Algorithms that parse data, learn from it, and make predictions without explicit rules.

Instead of hard-coding logic, we minimize a loss function over parameters ϕ:

$$\min_{\phi} \mathcal{L}(\phi) = \sum (y_{target} - f_{\phi}(x_{input}))^2$$

It is essentially high-dimensional curve fitting.

[IMG: meme_ml.jpg]
(When the loss finally converges.)

2. What is Reinforcement Learning (RL)?

ML with no "correct answer" data. Just an agent, an environment, and a **reward signal**.

The Goal: Maximize the expected return $G(\tau_\phi)$ over trajectories τ_ϕ.

$$J(\phi) = \mathbb{E}_{\tau_\phi \sim \pi_\phi}[G(\tau_\phi)]$$

The agent learns by trial and error (Sutton and Barto, 2018).

[IMG: meme_rl.jpg]

(The training process...)

3. What is a Policy (π)?

The Definition: A policy is the function that maps **States** $\rightarrow$ **Actions**.

Notation: We denote the policy for agent i as $\pi(a^i|s, \phi^i)$, parameterized by ϕ^i.

In Deep RL: The policy is a Neural Network.

$$a_t^i \sim \pi(\cdot|s_t, \phi^i)$$

Our goal is to find the parameters ϕ^i that maximize the agent's value function V^i.

[IMG: meme_policy.jpg]
(The policy decides: Explore or Exploit?)

4. Policy Gradient Methods

How do we optimize ϕ^i directly?

The Policy Gradient Theorem: We differentiate the probability of our trajectory τ_ϕ.

$$\nabla_{\phi^i} J(\phi^i) = \mathbb{E}\left[\nabla_{\phi^i} \log \pi(\tau_\phi | \phi^i) \cdot G^i(\tau_\phi)\right]$$

- We push the parameters ϕ^i in the direction that makes high-reward trajectories (G^i) more likely.

[IMG: meme_pg.jpg]
(Calculus to the rescue.)

5. What is Multi-Agent RL (MARL)?

The world is not empty. Peer agents exist (denoted as $-i$).

The Core Problem: Non-Stationarity. As peer agents update their parameters ϕ^{-i}, their policies change. From agent i's perspective, the **environment dynamics shift**.

$$P(s'|s, a^i, \pi_{\phi^{-i}}) \neq \text{const}$$

Standard RL assumptions break. We need specific Multi-Agent Policy Gradients.

[IMG: meme_marl.jpg]
(Who changed the environment? You did!)

6. Meta-Learning ("Learning to Learn")

Standard agents react. **Meta-Agents anticipate.**
A meta-learner i optimizes its initial parameters ϕ_0^i to maximize value over a **sequence of future updates** (a Markov chain of policies).

The Objective:

$$\max_{\phi_0^i} \mathbb{E}\left[\sum_{l=0}^{L} V^i_{\phi_{0:l+1}}(s_0, \phi_0^i)\right]$$

We optimize not just for the current step, but for the entire adaptation process.

[IMG: meme_meta.jpg]
(Optimizing the optimizer.)

7. The Meta-Multiagent Policy Gradient (Meta-MAPG)

Adopting notation from Kim et al. (2021), the gradient of the meta-value function decomposes into three terms:

$$\nabla_{\phi_0^i} V \approx \mathbb{E}\Bigg[G^i(\tau) \cdot \Bigg(\underbrace{\nabla_{\phi_0^i} \log \pi(\tau_{\phi_0} | \phi_0^i)}_{\textbf{Current Policy}} + \underbrace{\sum \nabla_{\phi_0^i} \log \pi(\tau | \phi^i_{\ell'+1})}_{\textbf{Own Learning}} + \underbrace{\sum \nabla_{\phi_0^i} \log \pi(\tau | \phi^{-i}_{\ell'+1})}_{\text{Peer Learning}} \Bigg) \Bigg]$$

[IMG: meme_metapg.jpg]

(Differentiating through the opponent's learning step.)

The Key Mechanism:

- The **Peer Learning** term (Eq. 4 in Kim et al.) is the "Collusion Term."
- It measures: *"How does my initial choice ϕ_0^i steer the opponent's future updates ϕ^{-i}?"*

8. Application: Algorithmic Collusion

Bertrand Competition: Algorithms competing for customers should drive prices to Marginal Cost ($P = MC$).

The Hypothesis: A Meta-Learner can exploit the **Peer Learning** term to "teach" a naive agent to collude.
It accepts short-term losses (punishing price cuts) to steer the opponent's parameters ϕ^{-i} toward a high-price equilibrium ($P \gg MC$).

[IMG: meme_collusion.jpg]
(Stonks.)

9. The Tech Stack

- **PyTorch:** Custom implementation of Meta-MAPG.
- **PettingZoo:** Multi-agent environment wrapper.
- **Cluster Compute:** Essential for calculating higher-order gradients ($\nabla_{\phi^i} \nabla_{\phi^{-i}}$).

Key Libraries:

`torch.autograd`
(For differentiating through the update step)

`wandb`
(Experiment tracking)

10. Implications for Alignment

- **Emergent Deception:** Collusion is a form of economic deception. We didn't program it; it emerged from meta-optimization.
- **The "Teacher" Risk:** As AI systems become more capable, they may learn to manipulate the learning process of less capable systems (or humans).
- **Mechanism Design Failure:** "Fair rules" are insufficient against agents that can steer the equilibrium.

11. Resources & Next Steps

Thank you!

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The Code

GitHub

github.com/Meugenn
(Coming Soon)

13. References I

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